

Software Engineering & Project Management Lab Experiment No: - 05

Aim: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Theory:

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily leading to multiple integrations per day. Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Start building your software project

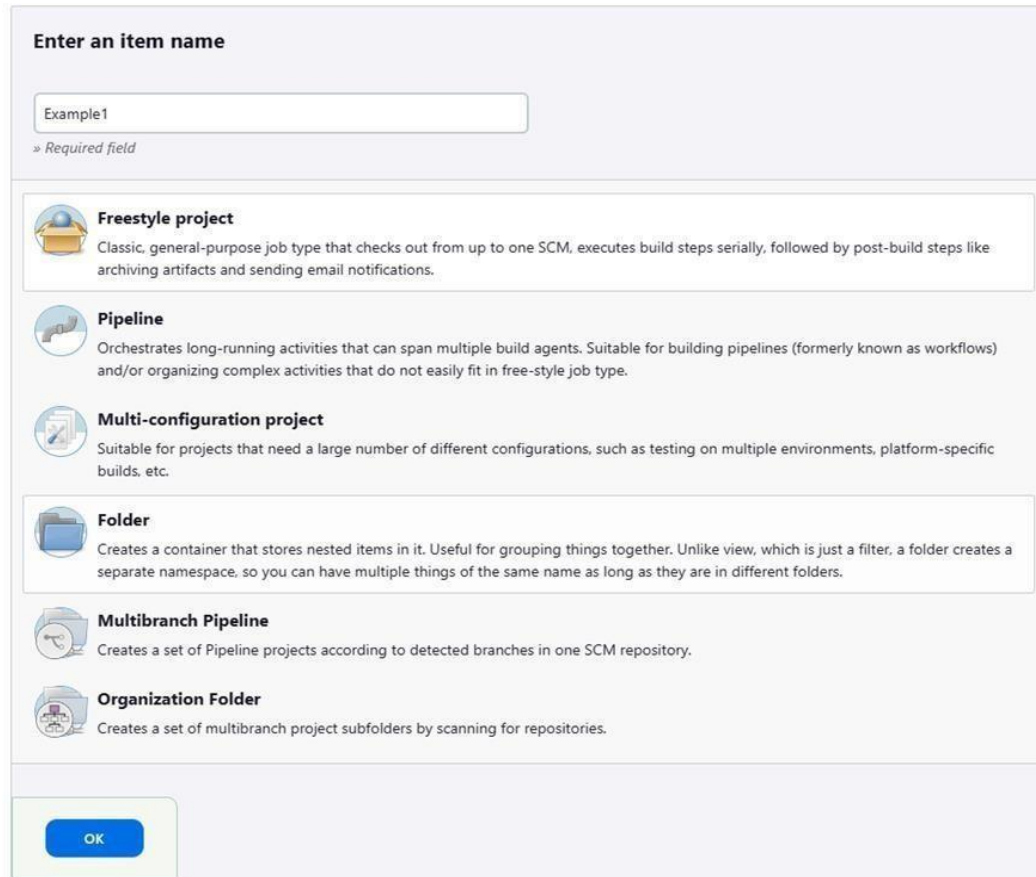
Create a job



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Naming the job and setting it as freestyle:



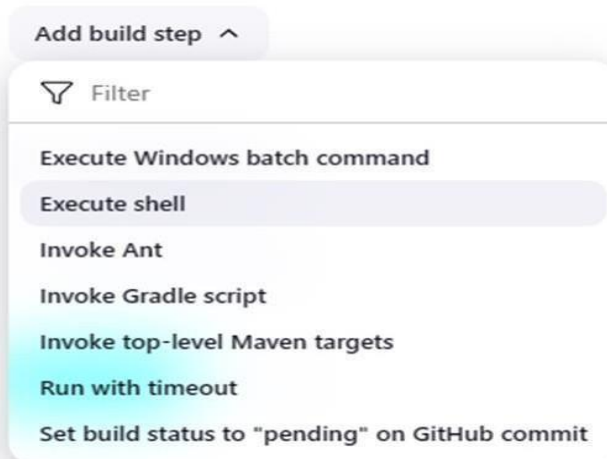
The image shows the 'Enter an item name' dialog in Jenkins. At the top, there is a text input field containing 'Example1' and a label '» Required field'. Below this, there is a list of job types, each with an icon and a description:

- Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**: Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**: Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**: Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**: Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organization Folder**: Creates a set of multibranch project subfolders by scanning for repositories.

At the bottom left of the dialog is a blue 'OK' button.

Selecting build type as “Execute shell”:

Build Steps



The image shows the 'Add build step' dropdown menu in Jenkins. The menu is open, showing a list of build steps. The first option is 'Execute shell', which is highlighted with a light blue background. Other options include 'Execute Windows batch command', 'Invoke Ant', 'Invoke Gradle script', 'Invoke top-level Maven targets', 'Run with timeout', and 'Set build status to "pending" on GitHub commit'.

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Entering a simple command for the shell execution:

Build Steps

≡ Execute shell ?

Command

See [the list of available environment variables](#)

```
echo "Hello TSEC"
```

Advanced ▾

Applying and saving the project configuration:

Save

Apply

✓ Saved

Building the project:

▶ Build Now

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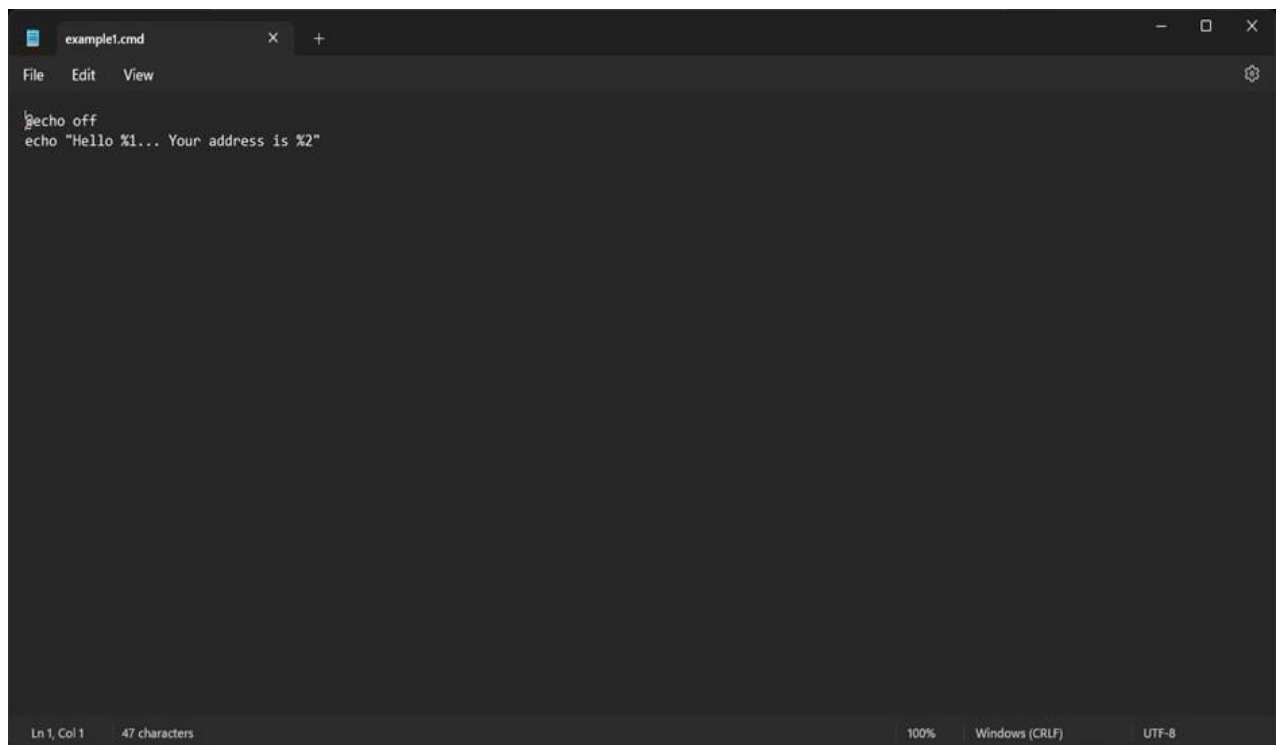
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Console output (after building):



Example 1.2: Taking parameters through files

Contents of script example1.cmd:



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Executing script example1.cmd on the terminal:

```
Microsoft Windows [Version 10.0.22621.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Users\AI&DS 202>Microsoft Windows [Version 10.0.22631.3155] (c) Microsoft Corporation. All rights reserved.
'Microsoft' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cmd
The system cannot find the path specified.

C:\Users\AI&DS 202>"Hello... Your address is "
"Hello... Your address is " is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cad Tanishq
The system cannot find the path specified.

C:\Users\AI&DS 202>"Hello Tanihsq... Your address is "
"Hello Tanihsq... Your address is " is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cmd Tanishq Girgaon "Helle Tanishq... Your address is Gi
rgaon"
The system cannot find the path specified.
```

Modifying the Jenkins project to execute the script while supplying required parameters:

Build Steps

The screenshot shows the Jenkins 'Execute Windows batch command' build step configuration. The 'Command' field contains the text: `C:\Admin\Academics\TSEC\Start3\SEPM\example1.cmd Siddhant Goregaon`. Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration box is an 'Add build step' button.

Console output after building the modified project:

The screenshot shows the Jenkins console output for a build. The status is 'SUCCESS'. The output text is as follows:

```
Started by user: Siddhant Chatur
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example1
[Example1] $ cd /c & call C:\WINDOWS\TEMP\jenkins70790983285501350.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example1>C:\Admin\Academics\TSEC\Start3\SEPM\example1.cmd Siddhant Goregaon
"Hello Siddhant... Your address is Goregaon"
Finished: SUCCESS
```

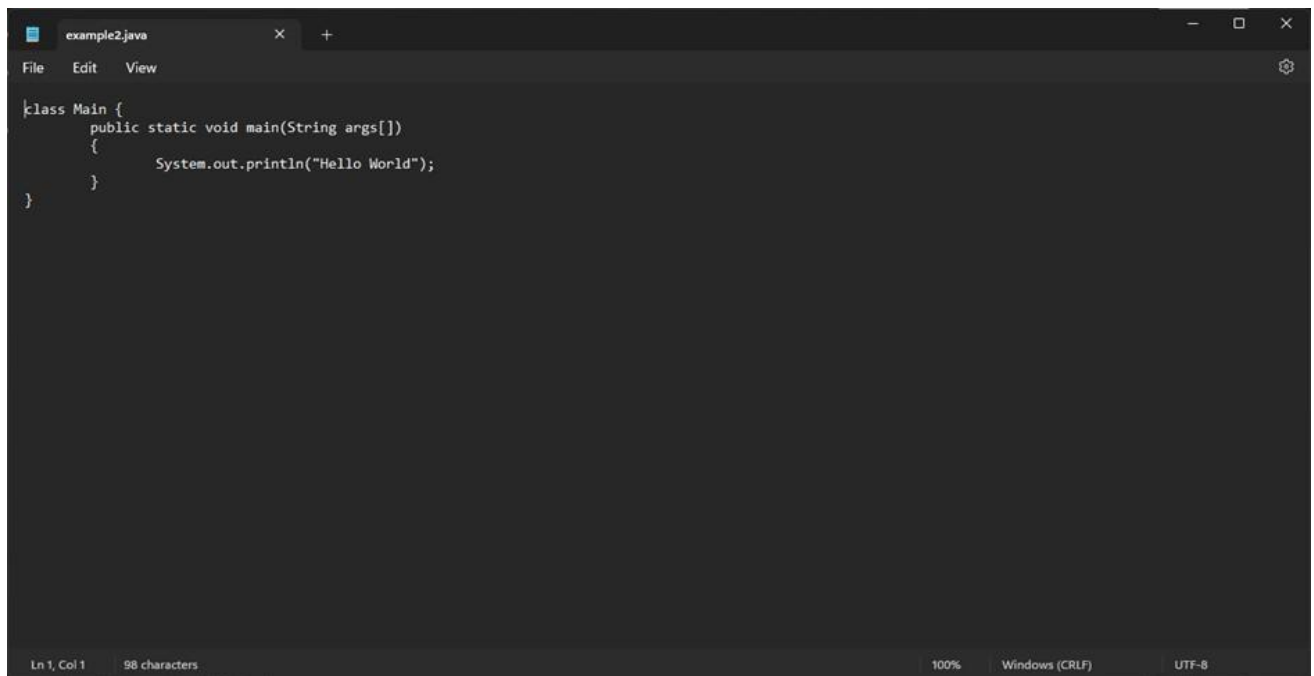
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Example 2

Example 2.1: Running a Java program under Jenkins

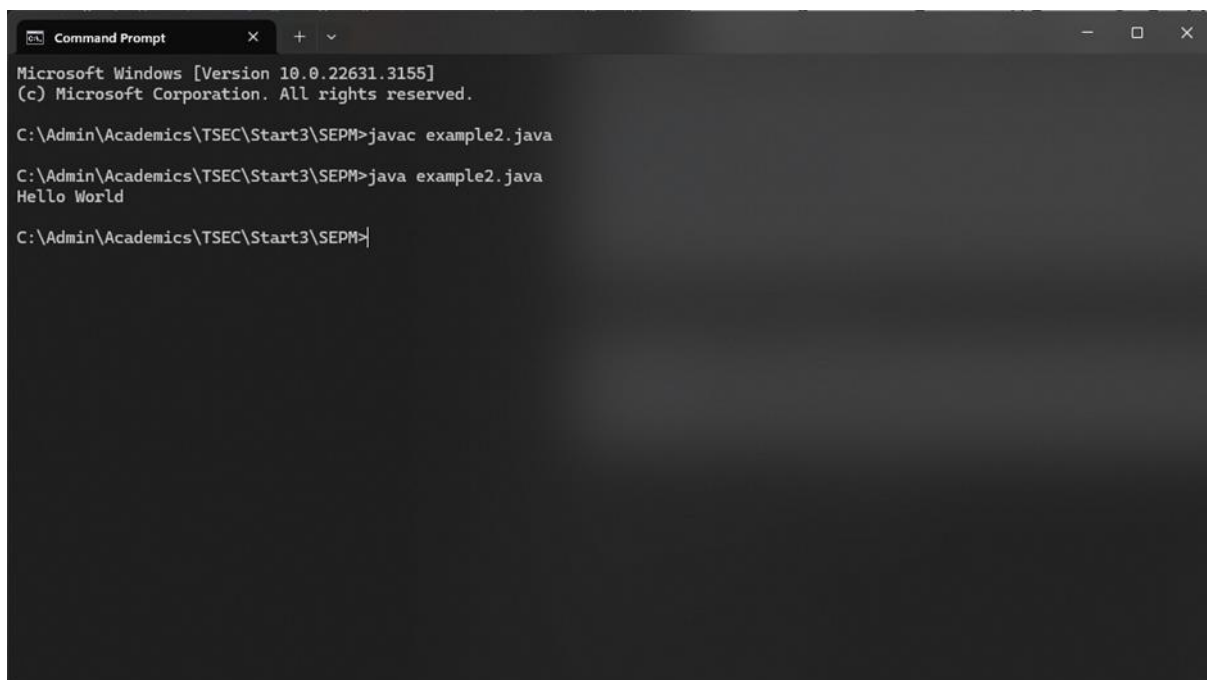
Creating a simple Java program:

A screenshot of an IDE window titled 'example2.java'. The code is as follows:

```
class Main {  
    public static void main(String args[])  
    {  
        System.out.println("Hello World");  
    }  
}
```

The status bar at the bottom indicates 'Ln 1, Col 1', '98 characters', '100%', 'Windows (CRLF)', and 'UTF-8'.

Compiling and running the program on the terminal:

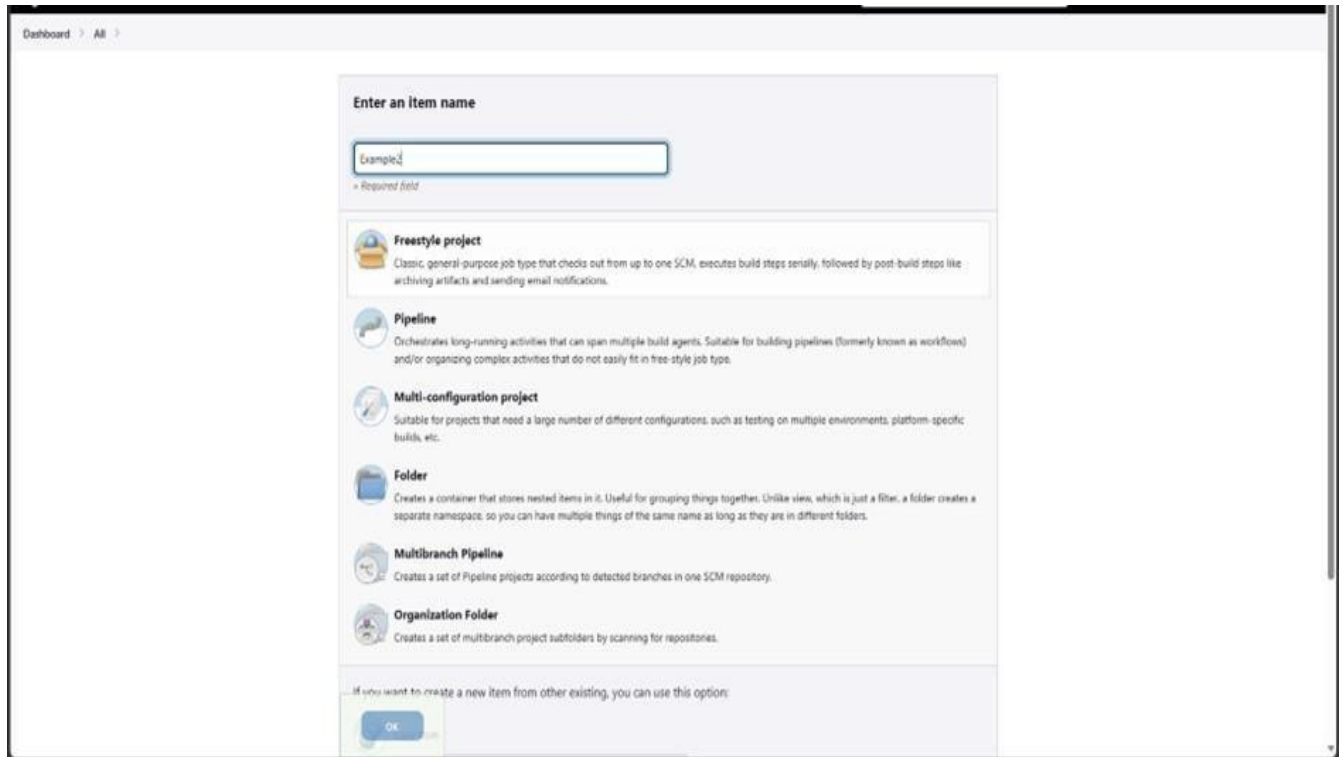
A screenshot of a Windows Command Prompt window. The text shown is:

```
Microsoft Windows [Version 10.0.22631.3155]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Admin\Academics\TSEC\Start3\SEPM>javac example2.java  
  
C:\Admin\Academics\TSEC\Start3\SEPM>java example2.java  
Hello World  
  
C:\Admin\Academics\TSEC\Start3\SEPM>|
```

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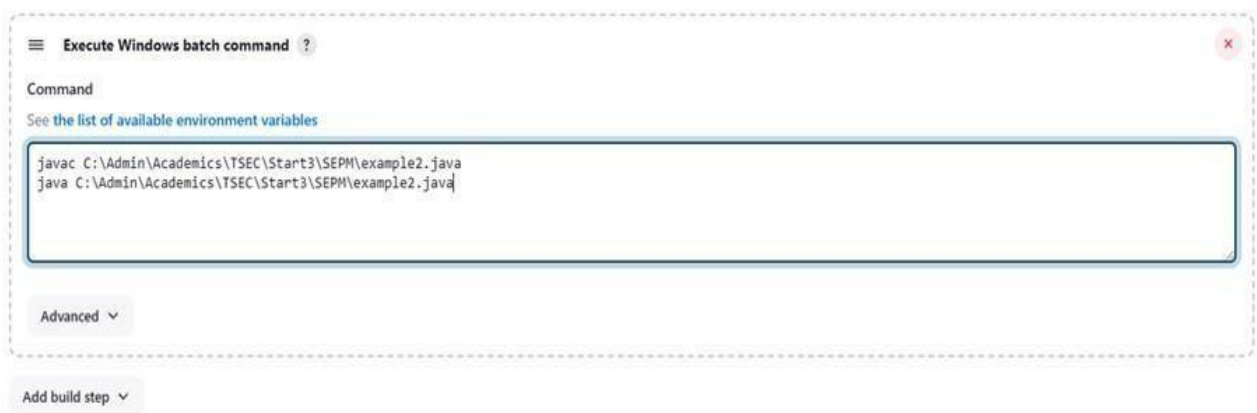
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Creating a new freestyle project:



Configure new project:

Build Steps



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Console output after building:

Console Output

```
Started by user Siddhant Chetlur
Running as SYSTEM
[EnvInject] - Loading node environment variables.
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example2
[Example2] $ cd /c call C:\WINDOWS\TEMP\jenkins15296462484398614135.bat

C:\ProgramData\Jenkins\jenkins\workspace\Example2>javac C:\Admin\Academics\TSEC\Start3\SEPM\example2.java

C:\ProgramData\Jenkins\jenkins\workspace\Example2>java C:\Admin\Academics\TSEC\Start3\SEPM\example2.java
Hello World

C:\ProgramData\Jenkins\jenkins\workspace\Example2>exit 0
Finished: SUCCESS
```

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enter an item name

» Required field

**Freestyle project**

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Pipeline**

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

**Multi-configuration project**

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

**Folder**

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

**Multibranch Pipeline**

Creates a set of Pipeline projects according to detected branches in one SCM repository.

**Organization Folder**

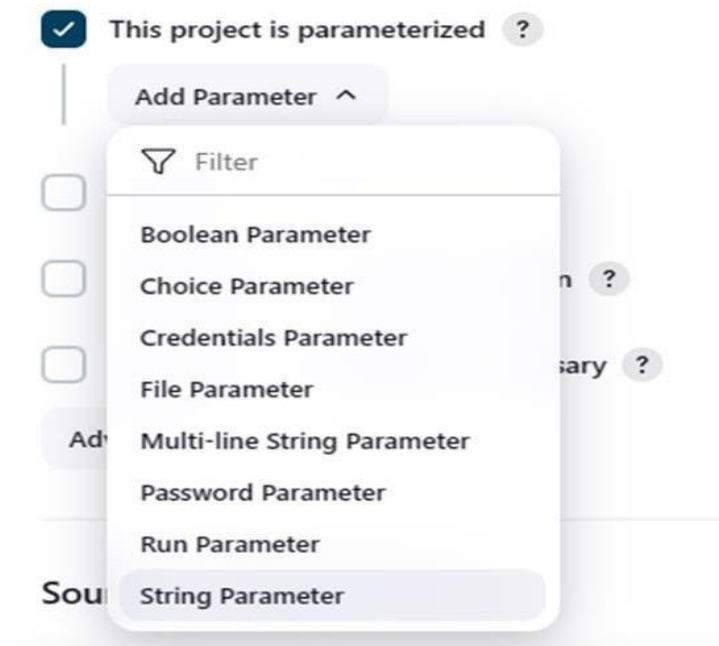
Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

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Enabling parameterisation and adding a String parameter:



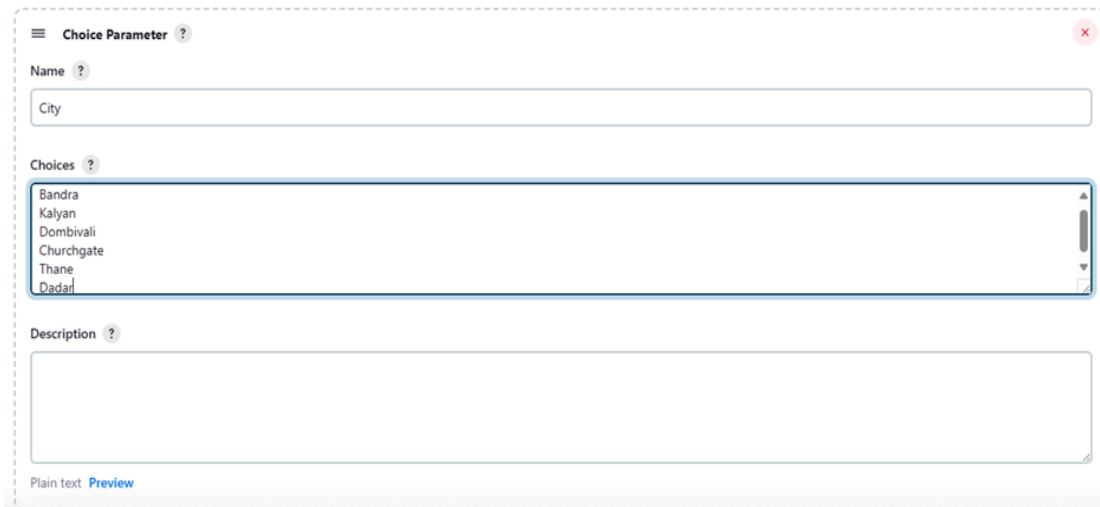
Configuring the string parameter as Fname:

A screenshot of the Jenkins 'String Parameter' configuration form. The form is titled 'String Parameter' with a help icon and a close button. It contains the following fields: 'Name' with the value 'Fname', 'Default Value' (empty), and 'Description' (empty). Below these fields is a 'Plain text' label and a 'Preview' link. At the bottom, there is a checkbox labeled 'Trim the string' which is currently unchecked.

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Adding a choice parameter and configuring it as City with the following choices:



The image shows the Jenkins 'Choice Parameter' configuration dialog. It has a title bar 'Choice Parameter' with a help icon. Below the title bar, there is a 'Name' field with a question mark icon, containing the text 'City'. Below that is a 'Choices' field with a question mark icon, containing a list of city names: 'Bandra', 'Kalyan', 'Dombivli', 'Churchgate', 'Thane', and 'Dadar'. Below the choices is a 'Description' field with a question mark icon, which is currently empty. At the bottom left, there is a 'Plain text' label and a 'Preview' link.

Creating a script which takes 2 arguments for name and city:

```
C:\Users\AI&DS 202>Microsoft Windows [Version 10.0.22631.3155] (c) Microsoft Corporation. All rights reserved.
'Microsoft' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPH>example3.cmd
The system cannot find the path specified.

C:\Users\AI&DS 202>Hello your name is and your city is
'Hello' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPH example3.cmd Tanishq
The system cannot find the path specified.

C:\Users\AI&DS 202>Hello your name is Tanishq and your city is
'Hello' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPH>example3.cmd Tansishq Bandra
The system cannot find the path specified.

C:\Users\AI&DS 202>Hello your name is Tanishq and your city is Bandra
'Hello' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPH|
```

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Configuring build steps:

Build Steps

Execute Windows batch command ?

Command

See [the list of available environment variables](#)

```
C:\Admin\Academics\TSEC\Start3\SEPH\example3.cmd %Fname% %City%
```

Advanced ▾

Add build step ▾

Entering parameters for build:

Project Example3

This build requires parameters:

Fname

Aditya

City

Bandra



Build

Cancel

Console output after building:

✓ Console Output

```
Started by user Siddhant Chetlur
Running as SYSTEM
[EnvInject] - Loading node environment variables.
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example3
[Example3] $ cmd /c call C:\WINDOWS\TEMP\jenkins14094536165150986151.bat

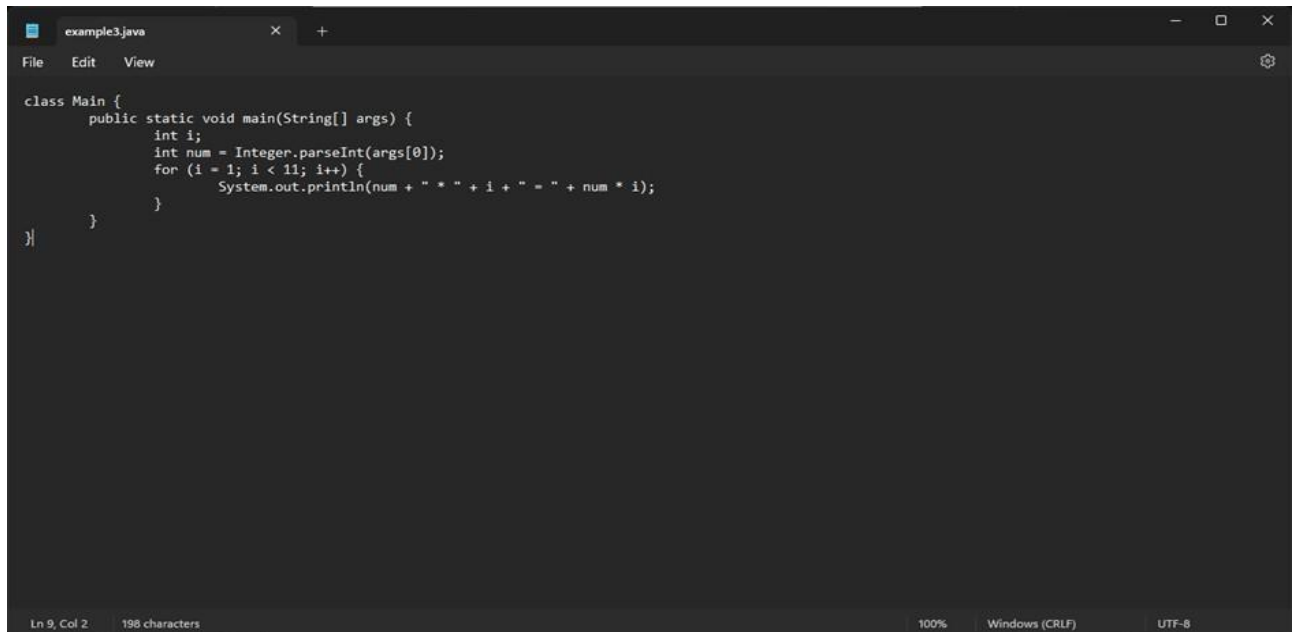
C:\ProgramData\Jenkins\jenkins\workspace\Example3>C:\Admin\Academics\TSEC\Start3\SEPH\example3.cmd Siddhant Bandra
Hello your name is Siddhant and your city is Bandra
Finished: SUCCESS
```

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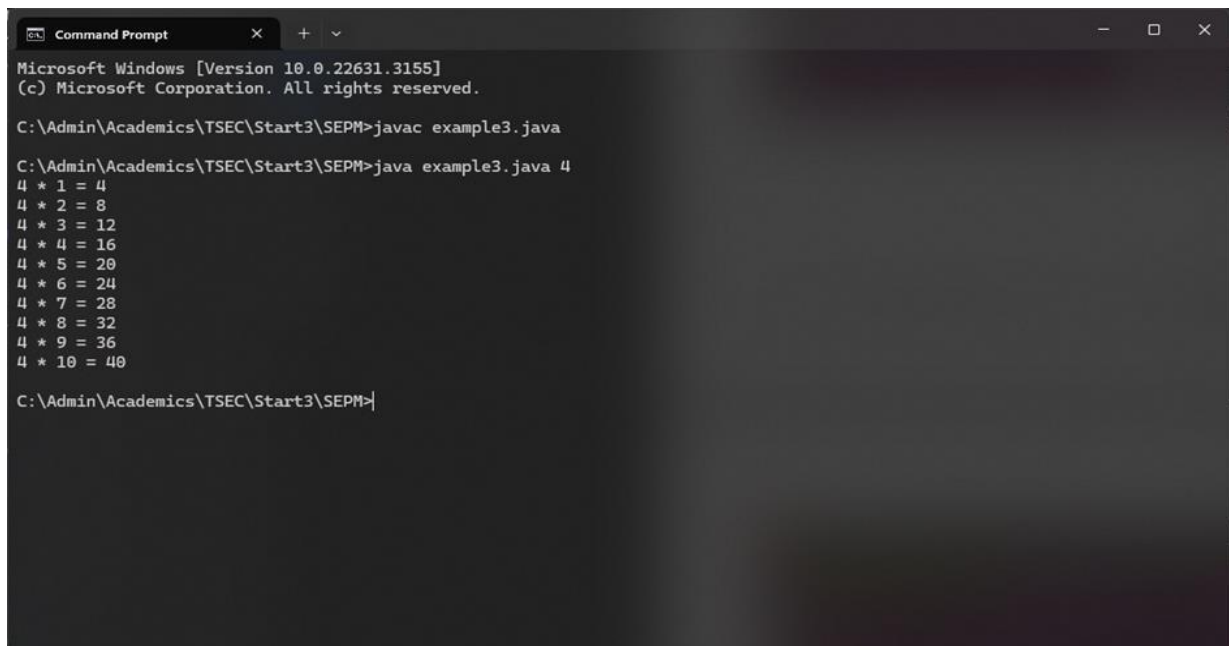
Example 3.2: Running a Java program with parameters

Creating a Java program with an input argument:



```
class Main {  
    public static void main(String[] args) {  
        int i;  
        int num = Integer.parseInt(args[0]);  
        for (i = 1; i < 11; i++) {  
            System.out.println(num + " * " + i + " = " + num * i);  
        }  
    }  
}
```

Testing the program on the terminal:



```
Microsoft Windows [Version 10.0.22631.3155]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Admin\Academics\TSEC\Start3\SEPM>javac example3.java  
  
C:\Admin\Academics\TSEC\Start3\SEPM>java example3.java 4  
4 * 1 = 4  
4 * 2 = 8  
4 * 3 = 12  
4 * 4 = 16  
4 * 5 = 20  
4 * 6 = 24  
4 * 7 = 28  
4 * 8 = 32  
4 * 9 = 36  
4 * 10 = 40  
  
C:\Admin\Academics\TSEC\Start3\SEPM>
```

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Creating a new freestyle project:

Enter an item name

Example4

» Required field

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

OK

Parameterise the project by adding a string parameter as follows:

☒ This project is parameterized ?

String Parameter ?

Name ?

num

Default Value ?

Description ?

Plain text [Preview](#)

☐ Trim the string ?

Add Parameter

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Configure the build steps:

Build Steps

Execute Windows batch command ?

Command

See [the list of available environment variables](#)

```
javac C:\Admin\Academics\TSEC\Start3\SEPM\example3.java
java C:\Admin\Academics\TSEC\Start3\SEPM\example3.java %num%
```

Advanced ▾

Add build step ▾

Entering the parameter for the build:

Project Example4

This build requires parameters:

num



Build

Cancel

Console output after building:

Console Output

```
Started by user Siddhant Chetlur
Running as SYSTEM
[EnvInject] - Loading node environment variables.
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example4
[Example4] $ cmd /c call C:\WINDOWS\TEMP\jenkins15119185770823247708.bat

C:\ProgramData\Jenkins\jenkins\workspace\Example4>javac C:\Admin\Academics\TSEC\Start3\SEPM\example3.java

C:\ProgramData\Jenkins\jenkins\workspace\Example4>java C:\Admin\Academics\TSEC\Start3\SEPM\example3.java 25
25 * 1 = 25
25 * 2 = 50
25 * 3 = 75
25 * 4 = 100
25 * 5 = 125
25 * 6 = 150
25 * 7 = 175
25 * 8 = 200
25 * 9 = 225
25 * 10 = 250

C:\ProgramData\Jenkins\jenkins\workspace\Example4>exit 0
Finished: SUCCESS
```

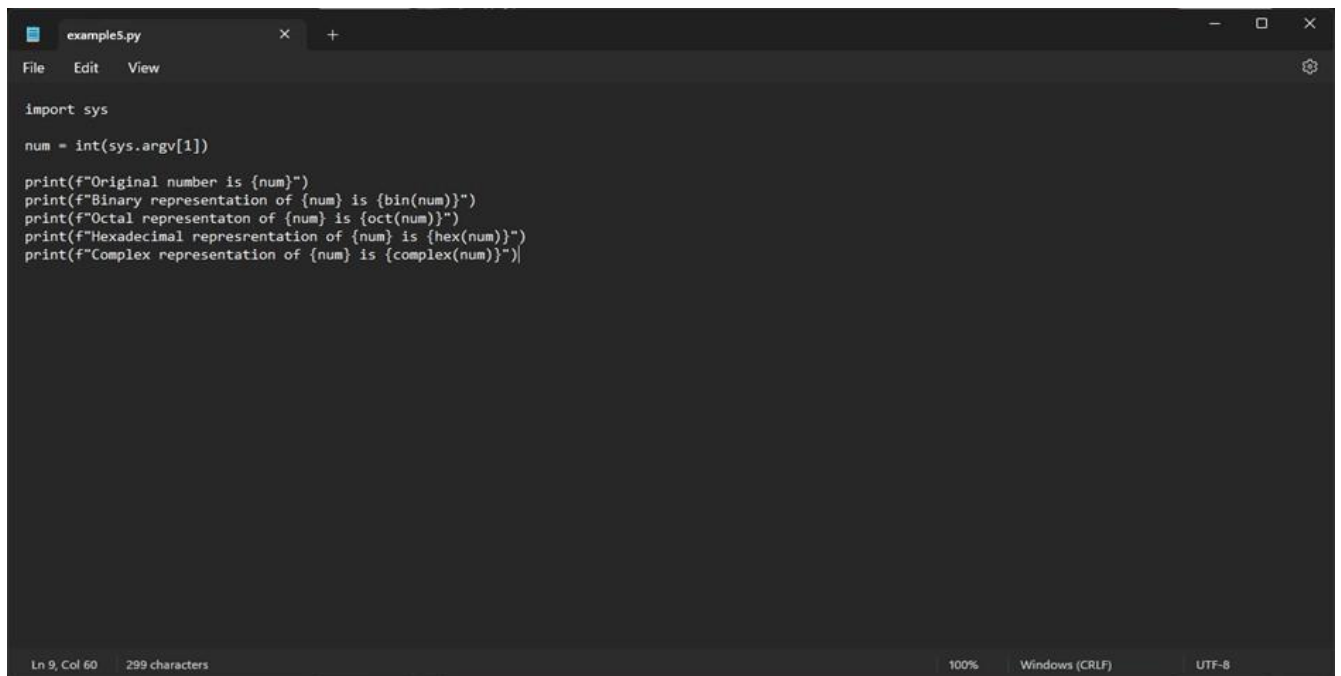
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Example 5

Example 5.1: Running a Python program

Creating a simple Python script:



```
example5.py
File Edit View

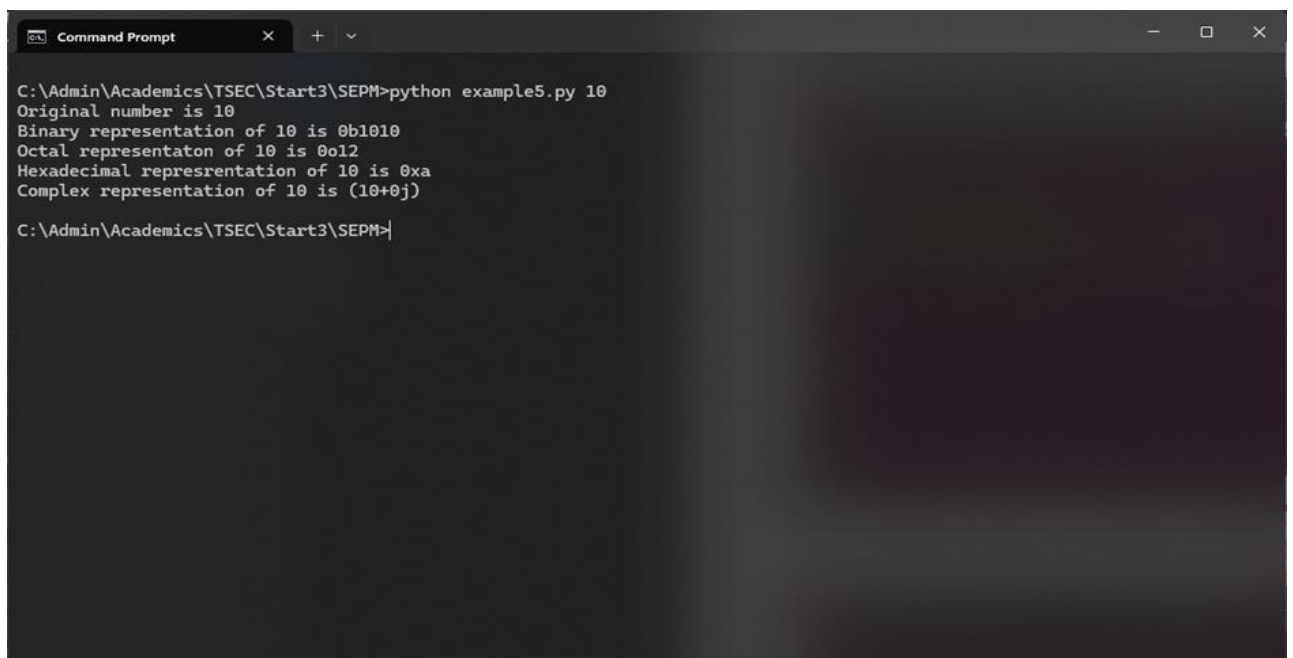
import sys

num = int(sys.argv[1])

print(f"Original number is {num}")
print(f"Binary representation of {num} is {bin(num)}")
print(f"Octal representation of {num} is {oct(num)}")
print(f"Hexadecimal representation of {num} is {hex(num)}")
print(f"Complex representation of {num} is {complex(num)}")

Ln 9, Col 60 299 characters 100% Windows (CRLF) UTF-8
```

Running the Python script on the terminal:



```
Command Prompt

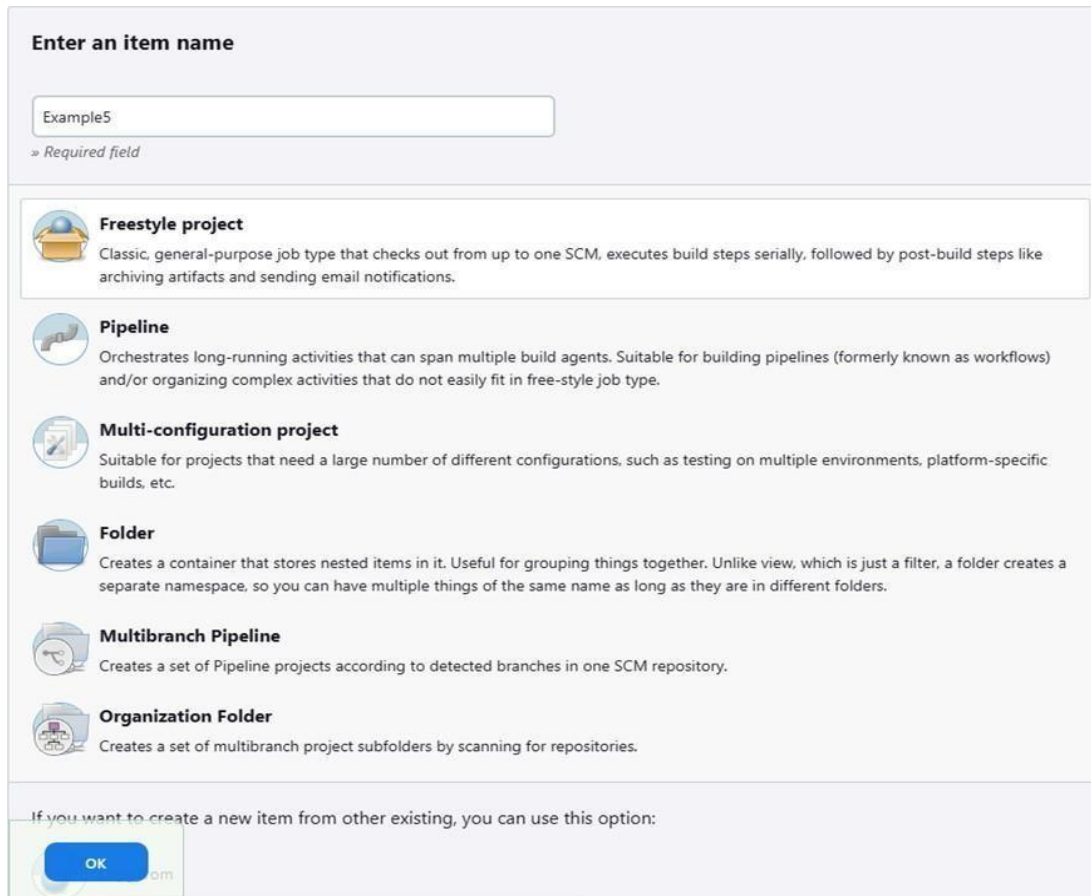
C:\Admin\Academics\TSEC\Start3\SEPM>python example5.py 10
Original number is 10
Binary representation of 10 is 0b1010
Octal representation of 10 is 0o12
Hexadecimal representation of 10 is 0xa
Complex representation of 10 is (10+0j)

C:\Admin\Academics\TSEC\Start3\SEPM>
```

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Creating a new freestyle project:



The screenshot shows the Jenkins 'Enter an item name' dialog. At the top, there is a text input field containing 'Example5' and a label '» Required field'. Below this, there is a list of project types, each with an icon and a description:

- Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**: Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
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- Folder**: Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**: Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organization Folder**: Creates a set of multibranch project subfolders by scanning for repositories.

At the bottom, there is a text label: 'If you want to create a new item from other existing, you can use this option:'. Below this label is a button labeled 'OK'.

Parameterising the project with a string parameter as follows:



The screenshot shows the Jenkins 'String Parameter' configuration dialog. At the top, there is a checkbox labeled 'This project is parameterized' which is checked. Below this, there is a section titled 'String Parameter' with a red 'X' icon in the top right corner. Inside this section, there are three input fields:

- Name**: A text input field containing 'num'.
- Default Value**: A text input field.
- Description**: A text area.

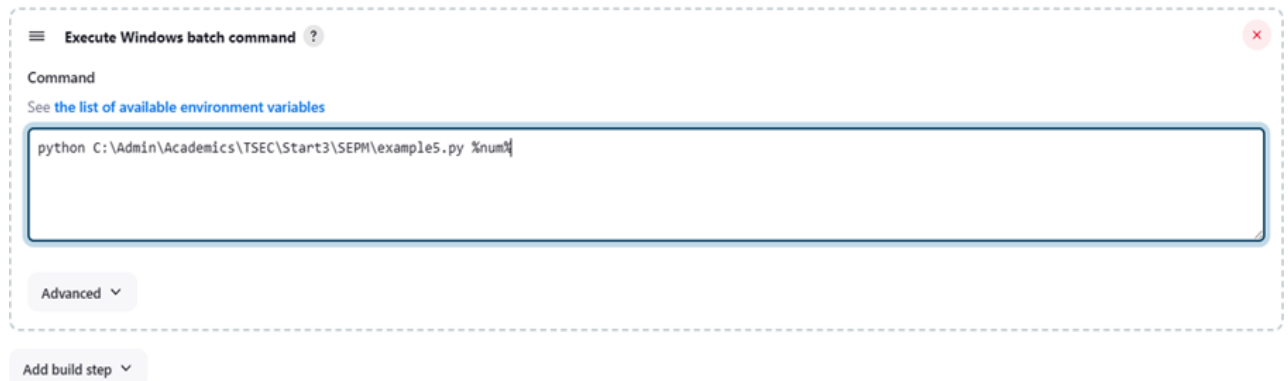
Below the input fields, there is a label 'Plain text' and a link 'Preview'. At the bottom, there is a checkbox labeled 'Trim the string' which is unchecked. Below the checkbox is a button labeled 'Add Parameter' with a dropdown arrow.

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Configuring the build steps:

Build Steps



The screenshot shows the Jenkins 'Execute Windows batch command' build step configuration. The title bar says 'Execute Windows batch command'. Below it, the 'Command' field contains the text: `python C:\Admin\Academics\TSEC\Start3\SEPM\example5.py %num%`. There is a link 'See the list of available environment variables' above the command field. Below the command field is an 'Advanced' dropdown menu. At the bottom, there is an 'Add build step' button.

Setting the parameter for the build:

Project Example5

This build requires parameters:

num

Console output after building:

Console Output

```
Started by user Siddhant Chetlur
Running as SYSTEM
[EnvInject] - Loading node environment variables.
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example5
[Example5] $ cmd /c call C:\WINDOWS\TEMP\jenkins11157306491994478222.bat

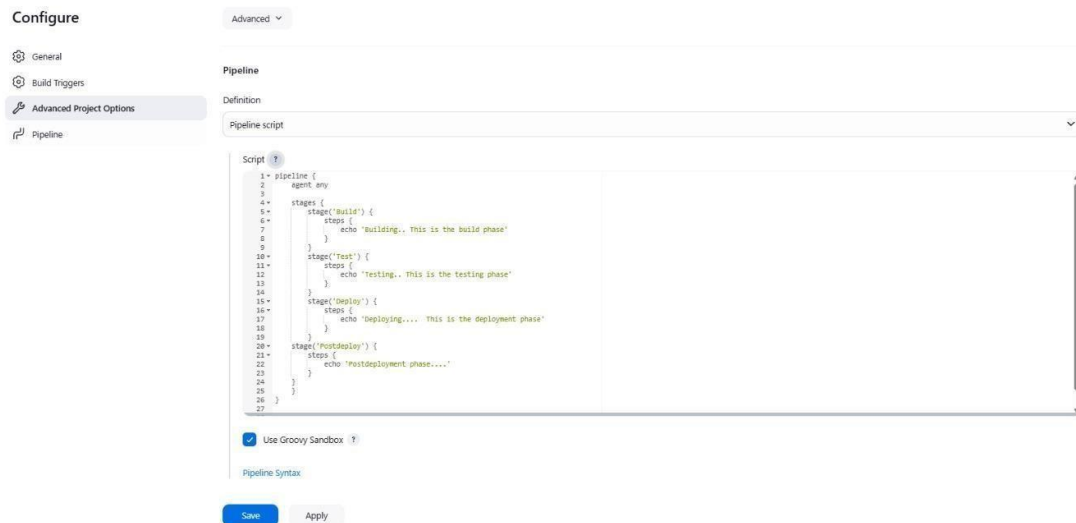
C:\ProgramData\Jenkins\jenkins\workspace\Example5>python C:\Admin\Academics\TSEC\Start3\SEPM\example5.py 10
Original number is 10
Binary representation of 10 is 0b1010
Octal representaton of 10 is 0o12
Hexadecimal representation of 10 is 0xa
Complex representation of 10 is (10+0j)

C:\ProgramData\Jenkins\jenkins\workspace\Example5>exit 0
Finished: SUCCESS
```

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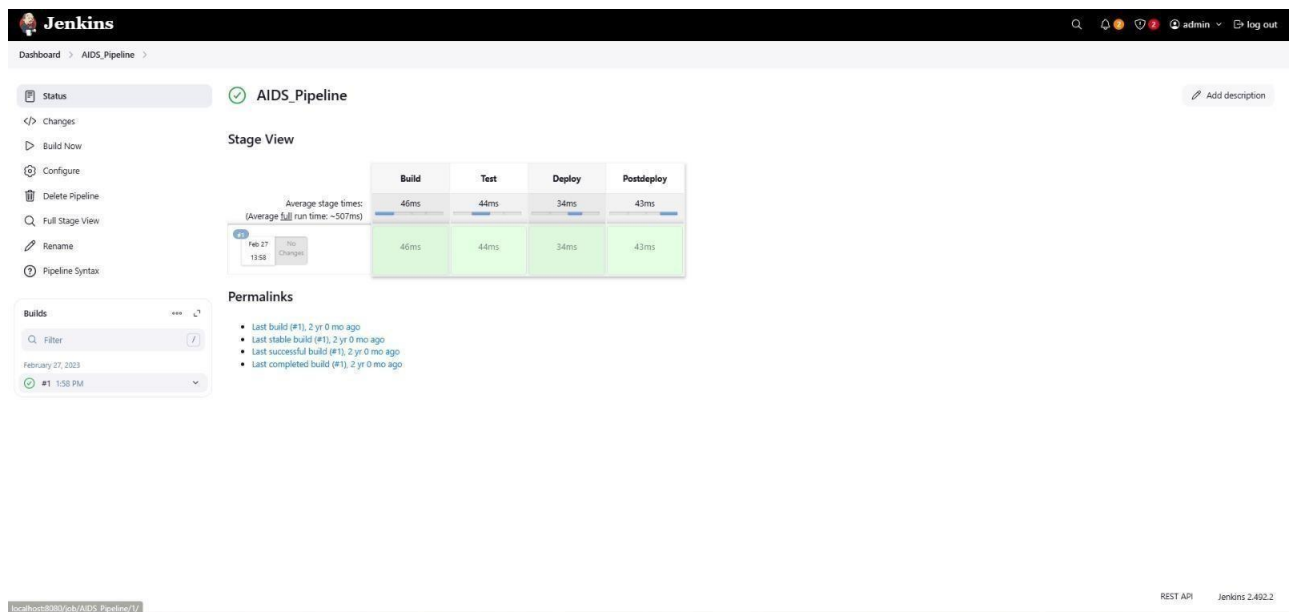
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Some Screenshots:



The screenshot shows the Jenkins 'Configure' page for a pipeline named 'AIDS_Pipeline'. The 'Pipeline' tab is selected, and the 'Definition' section shows a 'Pipeline script' dropdown. The script is a Groovy pipeline with four stages: 'Build', 'Test', 'Deploy', and 'Postdeploy'. Each stage has a single step with an 'echo' command. The 'Build' stage echoes 'Building.. This is the build phase', 'Test' echoes 'Testing.. This is the testing phase', 'Deploy' echoes 'Deploying.... This is the deployment phase', and 'Postdeploy' echoes 'Postdeployment phase....'. The 'Use Groovy Sandbox' checkbox is checked. 'Save' and 'Apply' buttons are at the bottom.

```
1 pipeline {
2   agent any
3
4   stages {
5     stage('Build') {
6       steps {
7         echo 'Building.. This is the build phase'
8       }
9     }
10    stage('Test') {
11      steps {
12        echo 'Testing.. This is the testing phase'
13      }
14    }
15    stage('Deploy') {
16      steps {
17        echo 'Deploying.... This is the deployment phase'
18      }
19    }
20    stage('Postdeploy') {
21      steps {
22        echo 'Postdeployment phase....'
23      }
24    }
25  }
26 }
27
```



The screenshot shows the Jenkins dashboard for the 'AIDS_Pipeline'. The 'Status' tab is selected, showing a green checkmark and 'Add description' button. The 'Stage View' section displays a table of stage execution times for the latest build (#1).

	Build	Test	Deploy	Postdeploy
Average stage times: (Average full run time: ~507ms)	46ms	44ms	34ms	43ms
Feb 17 13:58	46ms	44ms	34ms	43ms

The 'Permalinks' section lists recent build events:

- Last build (#1), 2 yr 0 mo ago
- Last stable build (#1), 2 yr 0 mo ago
- Last successful build (#1), 2 yr 0 mo ago
- Last completed build (#1), 2 yr 0 mo ago

The 'Builds' section shows a list of builds with a filter and a dropdown menu.

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Aim: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

The Jenkins Dashboard for the **AIDS_Pipeline** shows the following information:

- Stage View:** A table showing stage execution times for two builds.

	Build	Test	Deploy	Postdeploy
Average stage times: (Average full run time: ~1s)	103ms	48ms	50ms	54ms
#2 Mar 11 13:48	160ms	53ms	66ms	66ms
#1 Feb 27 13:58	46ms	44ms	34ms	43ms

- Permalinks:**
 - Last build (#1), 2 yr 0 mo ago
 - Last stable build (#1), 2 yr 0 mo ago
 - Last successful build (#1), 2 yr 0 mo ago
 - Last completed build (#1), 2 yr 0 mo ago
- Builds:** A list of builds with a filter and a table showing build status and time.

Build	Status	Time
#2	Success	1:45 PM
#1	Success	1:58 PM

The Jenkins Pipeline Steps view for the **AIDS_Pipeline** shows the following steps:

Step	Arguments	Status
Start of Pipeline - (2.1 sec in block)		Success
node - (0.78 sec in block)		Success
node block - (0.51 sec in block)		Success
stage - (0.19 sec in block)	Build	Success
stage block (Build) - (0.13 sec in block)		Success
echo - (13 ms in self)	Building. This is the build phase	Success
stage - (68 ms in block)	Test	Success
stage block (Test) - (39 ms in block)		Success
echo - (13 ms in self)	Testing. This is the testing phase	Success
stage - (80 ms in block)	Deploy	Success
stage block (Deploy) - (40 ms in block)		Success
echo - (1 ms in self)	Deploying... This is the deployment phase	Success
stage - (67 ms in block)	Postdeploy	Success
stage block (Postdeploy) - (40 ms in block)		Success
echo - (2 ms in self)	Postdeployment phase...	Success

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Aim: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

The screenshot displays the Jenkins interface for a pipeline named 'AIDS_Pipeline' at build #2. The 'Console Output' tab is active, showing the execution log. The pipeline starts with a 'Build' stage, followed by a 'Test' stage, and then a 'Deploy' stage. The 'Postdeploy' stage is also visible. The console output indicates that the pipeline is running on Jenkins in the directory C:\ProgramData\jenkins\workspace\AIDS_Pipeline. The 'Building.. This is the build phase' message is shown, followed by the 'Testing.. This is the testing phase' and 'Deploying.... This is the deployment phase' messages. The pipeline ends with 'Finished: SUCCESS'.

The 'Stage Logs (Build)' tab is also visible, showing the 'Building.. This is the build phase (self time 13ms)' message. Below this, the 'Stage View' table is displayed, showing the average stage times for the Build, Test, Deploy, and Postdeploy stages across three builds.

	Build	Test	Deploy	Postdeploy
Average stage times: (Average full run time: ~1s)	91ms	50ms	51ms	58ms
#3 Mar 11 13:47 No Changes	69ms	53ms	54ms	67ms
#2 Mar 11 13:46 No Changes	160ms	53ms	66ms	66ms
#1 Feb 27 13:58 No Changes	46ms	44ms	34ms	43ms

The 'Permalinks' section lists the last build (#3), last stable build (#2), last successful build (#2), and last completed build (#2), along with their respective timestamps.

Conclusion: Thus, we have successfully Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, created a pipeline script to Test and deploy an application over the tomcat server.